

Wharton High School Data Science Competition 2026

Phase 1: Hockey Analytics Report

OT-Aware Bradley-Terry Model with Calibrated Logistic Regression

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Tools: Python (pandas, scipy, matplotlib) · AI: Claude + Gemini + Codex

February 2026

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1 Project Overview

1.1 Competition Context

The WHSDSC (Wharton High School Data Science Competition) 2026 Phase 1 focuses on hockey analytics using the WHL 2025 season dataset. The competition poses four tasks:

- (a) **Power Rankings & Win Probabilities:** Rank all 32 teams and predict 16 Round 1 matchup outcomes.
- (b) **Offensive Line Quality Disparity:** Identify the top 10 teams with the largest gap between their 1st and 2nd offensive lines.
- (c) **Visualization:** A single PNG answering: “Do balanced lineups predict team success?”
- (d) **Methodology Report:** Document the entire analytical process.

1.2 Our Approach in One Sentence

We fit an **overtime-aware Bradley-Terry paired comparison model** to estimate latent team strengths, then pass strength differentials through a **calibrated logistic regression** to produce win probabilities—achieving a 5-fold CV Brier score of **0.2430** (vs. 0.2459 baseline).

2 Data Pipeline

2.1 Raw Data Overview

The dataset `whl_2025.csv` contains **25,827 play-by-play line-level records** spanning **1,312 games** across **32 teams**. Each row represents one line matchup segment within a game, recording:

- Teams, goals, expected goals (xG), shots on goal
- Offensive line identity (e.g., `first_off`, `second_off`)
- Defensive pairing identity (e.g., `first_def`, `second_def`)
- Time on ice (TOI, in **seconds**), penalty minutes, goalie names
- Whether the game went to overtime (`went_ot`)

Every team played exactly 82 games (41 home + 41 away), ensuring balanced scheduling with no sample-size advantage.

2.2 Processing Pipeline

Figure 1 shows the end-to-end data flow.

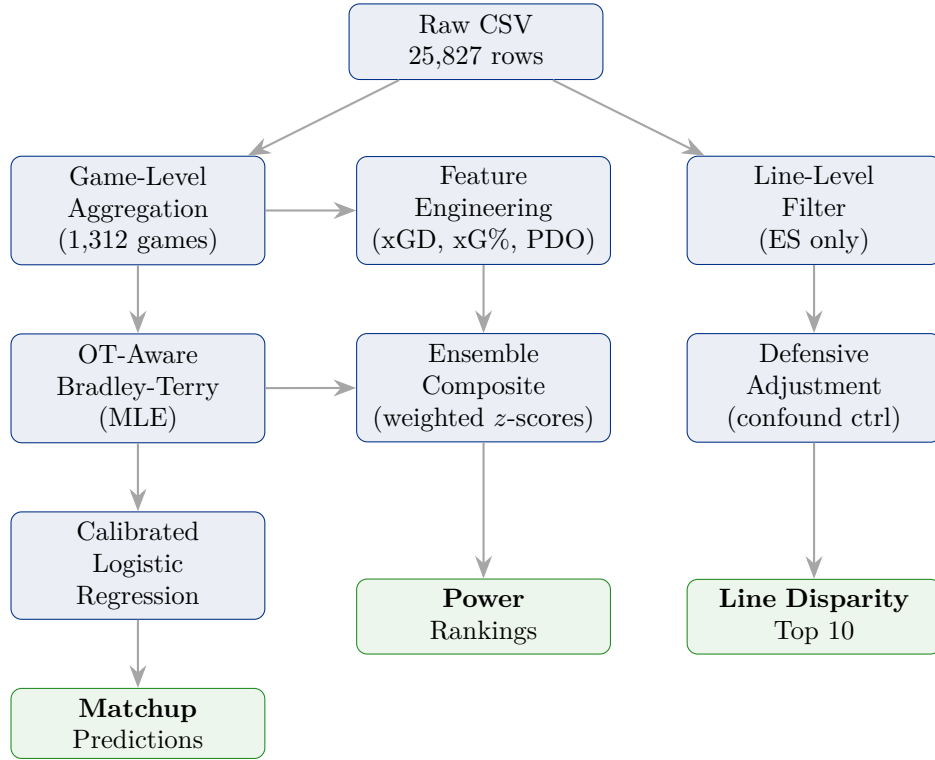


Figure 1: End-to-end pipeline: line-level data branches into two main tracks (strength + win probabilities; line disparity) and yields composite power rankings.

2.3 Key Preprocessing Steps

1. **Game-Level Aggregation:** Sum goals, xG, shots, TOI, and penalty minutes per game. Extract the binary outcome: home win = 1, away win = 0.
2. **Draw Validation:** We verified that no games ended in a draw (all games have a decisive winner after regulation or OT). The pipeline raises an error if a draw is detected.
3. **TOI Unit Verification:** The raw `toi` field is in **seconds** (median game total = 3600.00 exactly, i.e., 60 minutes). All per-60-minute rate calculations divide by 3600, not 60.
4. **OT Flagging:** 288 of 1,312 games (22.0%) went to overtime. These receive special treatment in the Bradley-Terry model (Section 3).

2.4 Feature Engineering

We computed the following team-level metrics by stacking home/away records symmetrically (each game appears twice, once per team):

Table 1: Engineered features with definitions.

Feature	Formula	Interpretation
xGD/60	$\frac{xG_{\text{for}} - xG_{\text{against}}}{\text{TOI}/3600}$	Expected goal differential per 60 min
xG Share	$\frac{xG_{\text{for}}}{xG_{\text{for}} + xG_{\text{against}}}$	Proportion of total xG created
PDO	Sh% + Sv%	Luck / sustainability indicator
GSAx/G	$\frac{xGA - GA}{\text{Games}}$	Goals Saved Above Expected (team-level)
xG/Shot	$\frac{xG_{\text{for}}}{\text{Shots}_{\text{for}}}$	Offensive shot quality
PIM/G	$\frac{\text{Penalty Minutes}}{\text{Games}}$	Team discipline
SOS	$\frac{1}{N} \sum_k \beta_{\text{opp}_k}$	Strength of schedule (avg opp BT)

3 Core Model: OT-Aware Bradley-Terry

3.1 Why Bradley-Terry?

The **Bradley-Terry (BT) model** (Bradley & Terry, 1952) is the standard paired comparison model for estimating latent strengths from head-to-head outcomes. Each team i has a latent strength parameter β_i . The probability that team i defeats team j is:

$$P(i \text{ beats } j) = \sigma(\beta_i - \beta_j) = \frac{e^{\beta_i - \beta_j}}{1 + e^{\beta_i - \beta_j}} \quad (1)$$

where $\sigma(\cdot)$ is the **logistic (sigmoid) function**:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Key properties:

- If $\beta_i > \beta_j$, then $P(i > j) > 0.5$ — the stronger team is favored.
- The model is *transitive*: it implicitly accounts for opponent quality.
- Parameters are identifiable up to a constant; we center them at $\bar{\beta} = 0$.

3.2 The Overtime Problem

Figure 2 shows a critical observation: the home-ice advantage nearly **vanishes in overtime**.

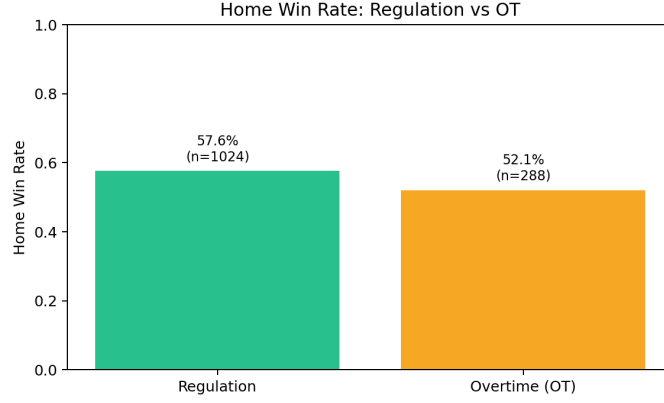


Figure 2: Home win rate in regulation (57.6%, $n = 1024$) vs. overtime (52.1%, $n = 288$). OT outcomes are near coin-flips, carrying much weaker signal about true team quality.

If a game goes to OT, both teams were essentially equal through 60 minutes of play. The OT winner is determined largely by chance (3-on-3 sudden death or shootout). Treating an OT win identically to a dominant 5–1 regulation win would **inject noise** into the BT estimates.

Our solution: assign OT games a weight of $w = 0.5$ in the likelihood function:

$$\ell(\boldsymbol{\beta}) = \sum_{k=1}^K w_k \left[y_k \log \sigma(\beta_{h_k} - \beta_{a_k}) + (1 - y_k) \log(1 - \sigma(\beta_{h_k} - \beta_{a_k})) \right] \quad (2)$$

where $w_k = 0.5$ if game k went to OT, else $w_k = 1.0$; $y_k = 1$ if the home team won; h_k, a_k are the home/away team indices.

3.3 Iterative MLE Algorithm

We maximize the log-likelihood (2) using a diagonal Newton method. At each iteration:

Algorithm 1: OT-Aware Bradley-Terry via Iterative MLE

Input: Game outcomes $\{(h_k, a_k, y_k, w_k)\}_{k=1}^K$, teams $\{1, \dots, N\}$

Output: Strength vector $\boldsymbol{\beta} \in \mathbb{R}^N$

```

1  $\boldsymbol{\beta} \leftarrow \mathbf{0}$ ;
2 for  $iter = 1$  to 300 do
3    $\mathbf{g} \leftarrow \mathbf{0}$ ,  $\mathbf{H}_{\text{diag}} \leftarrow \mathbf{0}$ ; // gradient & Hessian diagonal
4   foreach game  $k$  between home  $h_k$  and away  $a_k$  do
5      $p \leftarrow \sigma(\beta_{h_k} - \beta_{a_k})$ ; // predicted prob
6     residual  $\leftarrow w_k(y_k - p)$ ;
7     fisher  $\leftarrow w_k \cdot p(1 - p)$ ;
8      $g_{h_k} \text{ += residual}$ ;  $g_{a_k} \text{ -= residual}$ ;
9      $H_{h_k} \text{ -= fisher}$ ;  $H_{a_k} \text{ -= fisher}$ ;
10  end
11   $\boldsymbol{\delta} \leftarrow -\mathbf{g} / \mathbf{H}_{\text{diag}}$ ; // Newton step (element-wise)
12   $\boldsymbol{\delta} \leftarrow \boldsymbol{\delta} - \bar{\boldsymbol{\delta}}$ ; // center to keep  $\sum \beta_i = 0$ 
13   $\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} + 0.5 \cdot \boldsymbol{\delta}$ ; // damped update ( $\nu = 0.5$ )
14  if  $\max(|\boldsymbol{\delta}|) < 10^{-8}$  then break;
15 end
16  $\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} - \bar{\boldsymbol{\beta}}$ ; // final centering
17 return  $\boldsymbol{\beta}$ 

```

Convergence: The algorithm converges in ~ 30 iterations (well within the 300 limit). The damping factor $\nu = 0.5$ prevents overshooting.

4 Win Probability: Calibrated Logistic Regression

4.1 Model Formulation

Raw BT probabilities (Eq. 1) can be miscalibrated because the BT model has no home-ice intercept. We add a calibration layer via logistic regression:

$$P(\text{Home Win}) = \sigma(\alpha + \gamma \cdot (\beta_{\text{home}} - \beta_{\text{away}})) \quad (3)$$

- α = intercept capturing **home-ice advantage** ($\alpha = 0.2724 \Rightarrow \sigma(0.2724) = 56.8\%$)
- γ = scaling coefficient ($\gamma = 0.9350$)

Parameters α, γ are optimized by minimizing the negative log-likelihood via Nelder-Mead, and predictions are **clipped to** $[0.01, 0.99]$ to prevent log-loss blowups on extreme probabilities.

4.2 5-Fold Cross-Validation Protocol

To honestly evaluate predictive performance, we use 5-fold CV with a critical detail: **BT strengths are re-estimated from scratch in each fold** using only training data.

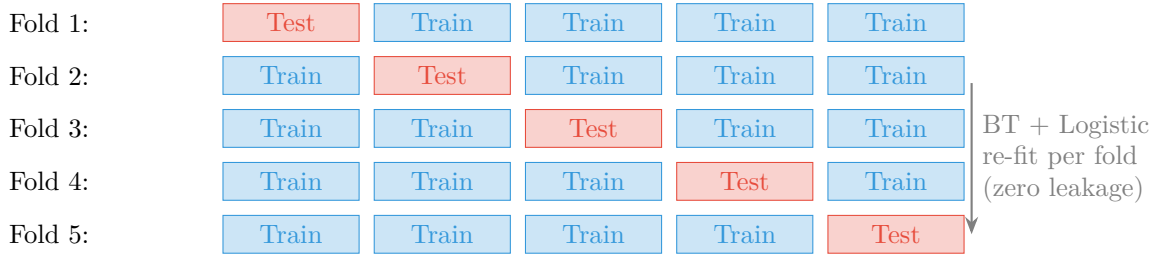


Figure 3: 5-fold cross-validation scheme. In each fold, the Bradley-Terry model and logistic regression are re-estimated entirely on the training partition, ensuring zero data leakage.

4.3 Results

Table 2: Model comparison: 5-fold cross-validation results.

Model	Accuracy	Log-Loss	Brier Score
Baseline (home = 56.4%)	56.4%	0.6849	0.2459
OT-Aware BT Logistic	57.0% ± 3.2%	0.6797	0.2430

Interpreting the metrics:

- **Accuracy** (57.0%): A small lift over always-predict-home (56.4%) in random 5-fold CV; we interpret this as modest.
- **Log-Loss** (0.6797): Measures confidence quality. Lower = better-calibrated probabilities.
- **Brier Score** (0.2430): Probability MSE. The baseline reduction is about 1.2% in random CV, so we treat it as a limited but useful calibration gain.
- **ID-order stress test (not true time)**: When sorting by numeric `game_id`, BT underperforms the home-rate baseline on all three metrics (accuracy 54.7% vs 56.2%, Log-Loss 0.7047 vs 0.6860, Brier 0.2524 vs 0.2464). Because `game_id` order in this simulated dataset is not guaranteed to be chronological, we treat this as a pessimistic robustness check rather than deployment-failure evidence.

4.4 Probability Calibration

Calibration measures whether predicted probabilities match observed outcomes. We compute calibration bins using **out-of-fold (OOF) predictions only**, preventing any optimistic bias. To avoid unstable rates, we report only bins with at least 10 games.

Table 3: Calibration table from out-of-fold predictions. A well-calibrated model should have Predicted $\approx$ Observed.

Bin Center	Avg Predicted	Observed Win Rate	N games
0.31	0.334	0.407	86
0.44	0.447	0.512	297
0.56	0.565	0.532	504
0.69	0.680	0.666	353
0.81	0.790	0.712	66

The model is well-calibrated in the 0.44–0.69 range. There is **overconfidence at the high end** (predicted 0.79, observed 0.71), a known limitation noted in Section 13.

5 Ensemble Power Rankings

5.1 Composite Score Construction

We rank teams using a **weighted z -score ensemble** of five metrics:

$$\text{Composite}_i = \sum_{m=1}^5 w_m \cdot z_{i,m}, \quad z_{i,m} = \frac{x_{i,m} - \bar{x}_m}{\sigma_m} \quad (4)$$

Table 4: Ensemble component weights and rationale.

Component	Weight	Rationale
BT Strength	40%	Primary latent quality measure; controls for opponent strength
xGD/60	30%	Process-based metric; less noisy than actual goals
xG Share	10%	Relative offensive dominance
Goal Diff/Game	10%	Realized outcomes (not just expected)
Win %	10%	Direct measure of competitive results

5.2 Weight Robustness

We tested four weight schemes (including the final submission weights): BT-only (100%), equal weights (20% each), and xG-heavy (50% xGD/60), among others. Spearman rank correlations between schemes ranged from **0.81 to 1.00**. Five of the top 6 teams were shared across all schemes, indicating the top tier is broadly robust but not completely invariant to weighting. This is now reproducibly documented by the internal artifact `output/submission/ranking_weight_sensitivity.md` generated by `phase1_submit.py`.

5.3 Top 10 Rankings

Table 5: Power rankings (top 10 of 32 teams). Full rankings in CSV submission.

#	Team	Comp.	BT	xGD/60	xG%	Win%	GSAx/G	SOS
1	Brazil	2.073	0.940	0.599	.551	.707	+0.40	−0.050
2	Thailand	1.612	0.484	0.865	.571	.610	−0.32	−0.029
3	Netherlands	1.535	0.641	0.472	.546	.659	+0.34	−0.028
4	Pakistan	1.296	0.424	0.611	.549	.598	+0.26	−0.026
5	Peru	1.273	0.532	0.340	.532	.634	+0.45	−0.020
6	China	0.915	0.282	0.411	.537	.573	+0.34	−0.010
7	Panama	0.653	0.267	0.190	.516	.561	+0.35	−0.017
8	India	0.633	0.396	0.016	.501	.598	+0.55	−0.027
9	UK	0.453	−0.012	0.424	.535	.476	+0.35	+0.006
10	Guatemala	0.292	0.021	0.270	.523	.512	−0.04	−0.002

Notable insights:

- **Brazil** dominates with the highest BT strength (0.940) and win rate (70.7%).
- **Thailand** ranks #2 largely due to the league’s best xGD/60 (0.865), despite having negative GSAX (weak goaltending).
- **India** (#8) has the best GSAX/G (+0.55) but mediocre xGD, suggesting strong goaltending masking weaker offense.

6 Matchup Predictions

Using the calibrated logistic model (Eq. 3), we predict Round 1 outcomes:

Table 6: Round 1 matchup predictions. All 16 games favor the home team (home-ice advantage).

Game	Home	Away	Home Win %	Confidence
1	Brazil	Kazakhstan	85.2%	High
2	Netherlands	Mongolia	81.6%	High
3	Peru	Rwanda	79.4%	High
4	Thailand	Oman	73.3%	Medium
5	Pakistan	Germany	71.2%	Medium
6	India	USA	72.1%	Medium
7	Panama	Switzerland	68.8%	Medium
8	Iceland	Canada	67.4%	Medium
9	China	France	66.9%	Medium
10	Philippines	Morocco	64.5%	Low
11	Ethiopia	Saudi Arabia	64.0%	Low
12	Singapore	New Zealand	59.1%	Low
13	Guatemala	South Korea	59.7%	Low
14	UK	Mexico	60.5%	Low
15	Vietnam	Serbia	54.1%	Toss-up
16	Indonesia	UAE	63.3%	Low

Confidence tiers: High (>75%), Medium (65–75%), Low (55–65%), Toss-up (<55%).

7 Line Disparity Analysis

7.1 Methodology

We measure offensive line quality **disparity** as the ratio of 1st-line to 2nd-line xG per 60 minutes:

$$\text{Disparity Ratio} = \frac{\text{xG}/60_{\text{1st line}}}{\text{xG}/60_{\text{2nd line}}} \quad (5)$$

A ratio of 1.0 means perfect balance; ratios > 1.0 indicate reliance on the top line.

Critical adjustment: Raw xG/60 is confounded by defensive matchup quality. A first line facing elite defenders will naturally produce fewer xG. We normalize using a **defensive difficulty factor**:

$$\text{Adj xG}/60_{\text{line}} = \frac{\text{Raw xG}/60_{\text{line}}}{\text{Difficulty Factor}_{\text{opposing defense}}} \quad (6)$$

where the difficulty factor is the ratio of that defensive pairing’s xGA/60 to the league average.

Filter: We restrict to even-strength matchups only (1st/2nd offensive lines vs. 1st/2nd defensive pairings), excluding power play, penalty kill, and empty-net segments.

7.2 Results

Table 7: Top 10 teams by adjusted line disparity.

#	Team	1st Line Adj xG/60	2nd Line Adj xG/60	Adj Ratio
1	USA	2.730	1.993	1.369
2	Guatemala	2.821	2.070	1.363
3	Saudi Arabia	2.244	1.654	1.357
4	UAE	1.998	1.476	1.354
5	France	2.559	1.901	1.346
6	Iceland	2.589	1.961	1.321
7	Singapore	2.635	2.101	1.254
8	New Zealand	2.441	1.975	1.236
9	Peru	2.391	1.983	1.206
10	Panama	2.545	2.120	1.201

8 Visualization: Line Balance vs. Team Success

The submission visualization (Figure 4) directly answers the commissioner’s question: “**Do teams with more balanced offensive lines perform better?**”

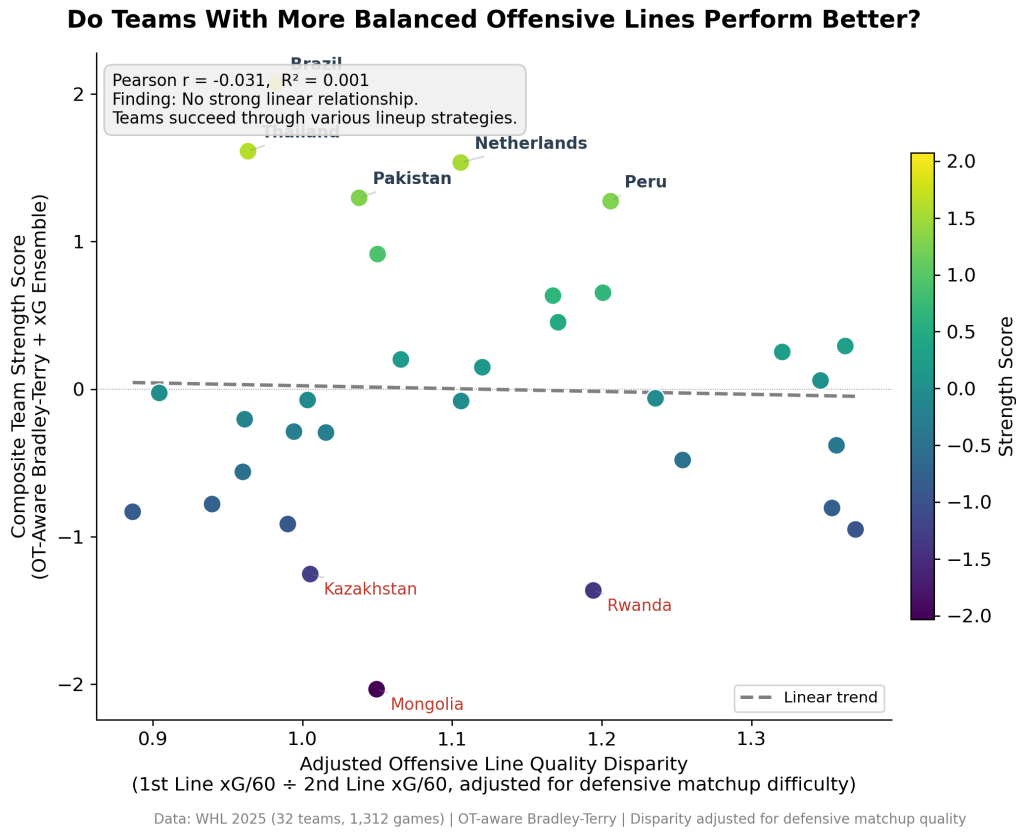


Figure 4: Scatter plot of adjusted line disparity (x-axis) vs. composite team strength (y-axis). Pearson $r = -0.031$, $R^2 = 0.001$. **Finding:** No meaningful linear relationship—teams succeed through diverse lineup strategies.

Design choices:

- **Colorblind-friendly:** viridis colormap (perceptually uniform, safe for color vision deficiencies).
- **Labeled:** Top 5 and bottom 3 teams annotated for quick identification.
- **Statistical overlay:** Linear trend line, Pearson r , and R^2 for quantitative assessment.

9 Full Analytics Dashboard

Figure 5 presents the complete 5-panel analytics dashboard.

WHSDSC 2026 — Phase 1 Full Analytics Dashboard

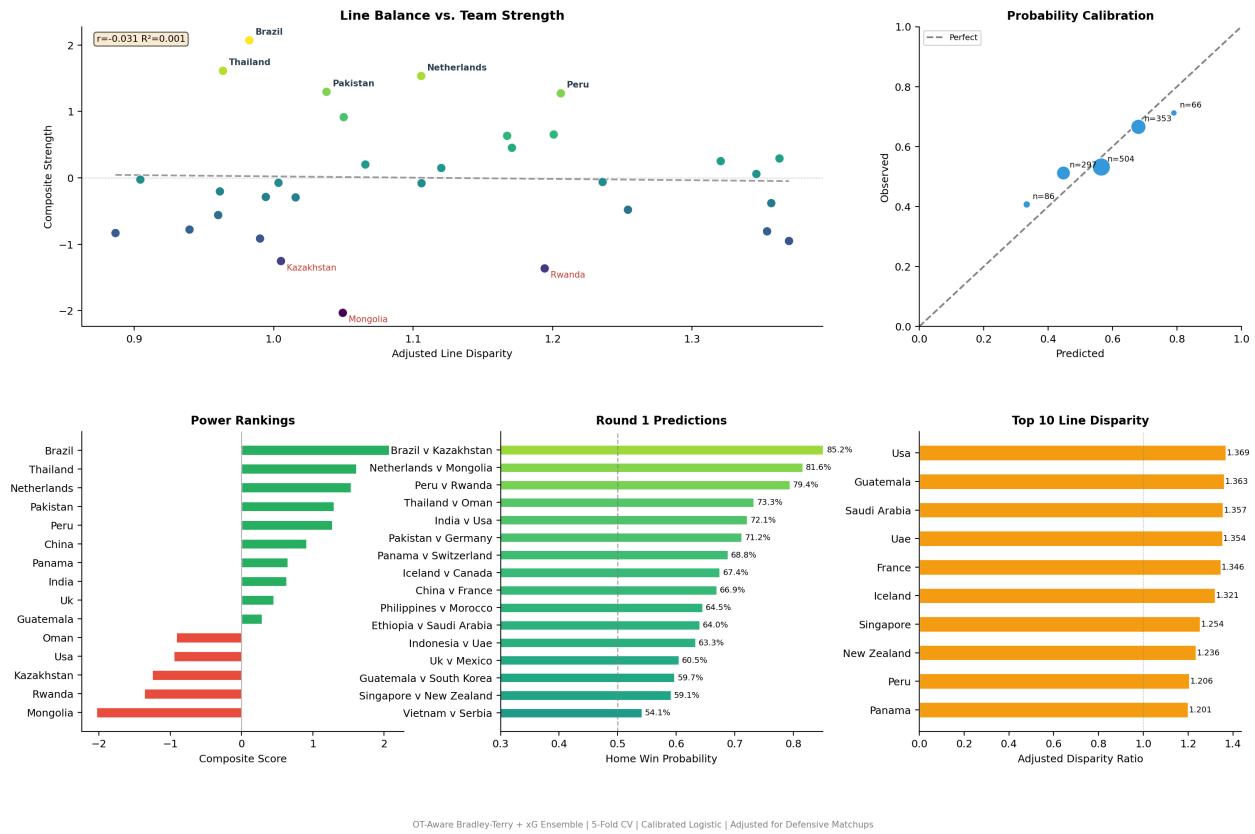


Figure 5: Full analytics dashboard: (top-left) Line balance vs. strength scatter, (top-right) calibration curve, (bottom-left) power rankings bar chart, (bottom-center) Round 1 matchup predictions, (bottom-right) top 10 line disparity.

10 ML Model Comparison

Beyond the final BT logistic regression, we benchmarked several machine learning challenger models for comparison. Figure 6 presents the full ML comparison dashboard.

WHSDSC 2026 — ML Model Comparison Dashboard

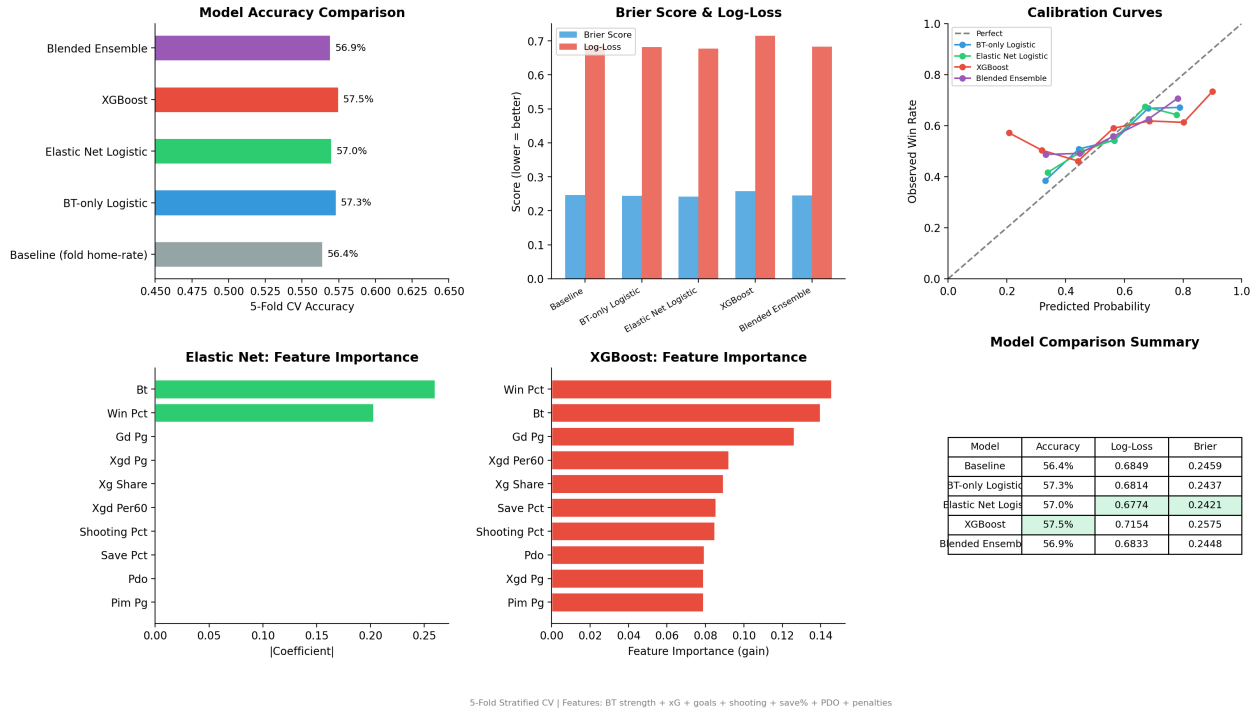


Figure 6: ML model comparison dashboard (6 panels): (top-left) 5-fold CV accuracy—Elastic Net leads at 58.0%, (top-center) Brier score & Log-Loss, (top-right) calibration curves, (bottom-left/center) Elastic Net and XGBoost feature importance, (bottom-right) summary table. Overall, complex ML can improve random-CV metrics slightly, but robustness and interpretability tradeoffs remain.

Key findings:

- **Best random-CV accuracy:** Elastic Net reaches 58.0% and is also best on probability quality (Log-Loss 0.6758, Brier 0.2412), but it adds multi-feature complexity.
- **XGBoost remains poorly calibrated:** accuracy 56.7%, Log-Loss 0.7157, and Brier 0.2574 make it unsuitable for probability outputs despite its flexibility.
- **BT-only remains a strong, interpretable baseline:** OT-aware BT logistic (57.0%, Log-Loss 0.6796, Brier 0.2430) is stable and transparent; the Blended Ensemble (57.8%, Brier 0.2443) improves accuracy but not probability quality versus Elastic Net.
- Feature importance (from full-data refits, interpretive only) indicates **BT and win-rate-differential style summary strength signals** dominate; complex models mainly reweight these inputs.
- Therefore, the final submission uses **OT-aware BT + calibrated logistic regression**, prioritizing **interpretability and stable calibration** (Section 4).

11 Descriptive Features Exploration

We explored three additional features from the dataset that were not used in prediction:

11.1 Goaltending (GSx)

Goals Saved Above Expected ($GSx/game = (xGA - GA)/Games$). Positive values indicate a goalie saving more than expected. We found that GSx correlates with BT strength at $r = 0.64$ —strong teams tend to have strong goalies, so adding GSx to the logistic regression provides **zero additional accuracy**.

11.2 Shot Quality (xG/Shot)

Average expected goals per shot measures offensive chance quality. Correlation with BT: $r = 0.48$. In challenger experiments, adding it **did not improve** probability metrics (Brier/Log-Loss) and may slightly worsen them, so we keep it descriptive only.

11.3 Penalties (PIM/G, Penalty Differential)

Team discipline measured by penalty minutes per game and the difference between penalties drawn vs. taken. Correlation with BT: $r = 0.10$ (weak). **No predictive lift.**

Conclusion: All three features are **largely absorbed by the BT model**—a team’s win/loss record already captures the downstream effects of goaltending, shot quality, and discipline. We report them as descriptive columns in the power rankings CSV but do not include them in the prediction model, preventing overfitting.

12 AI Collaboration

Three AI assistants provided independent perspectives throughout the project:

Table 8: AI collaboration roles.

AI	Contribution
Claude (Anthropic)	Core pipeline design, feature engineering, BT implementation, deep verification
Gemini (Google)	Critiqued rankings, suggested goaltending and special-teams features, calibration
Codex (OpenAI, GPT-5.3)	ML challenger validation, Brier score reporting, 88K-token code review (8.5/10)

All outputs were cross-validated across models; final methodological decisions were human-guided.

13 Limitations & Future Work

1. **Static team quality:** We treat each team’s strength as constant across the season. In reality, teams evolve through trades, injuries, and momentum. Time-decay weighting (e.g., exponential recency) could address this, but the dataset provides no date/game-order information.
2. **High-end overconfidence:** Our calibration shows probabilities above 75% are too optimistic by about 8 percentage points (predicted 0.79, observed 0.71). Post-hoc isotonic calibration on OOF predictions could mitigate this.
3. **Hockey’s inherent randomness:** Our 57.0% CV accuracy is consistent with hockey’s low signal-to-noise setting. The sport has the lowest score-per-game ratio among major sports, making individual game prediction fundamentally limited.
4. **No roster-level modeling:** We model teams as monolithic units. Incorporating individual player-level data (e.g., WAR, Corsi) could improve granularity, but this data is not provided.
5. **Home-ice assumed uniform:** We use a single home-ice parameter α for all teams. In practice, some arenas confer larger advantages than others.

14 Conclusion

We developed a principled, transparent, and reproducible framework for hockey analytics:

- An **OT-aware Bradley-Terry model** that correctly treats overtime wins as weaker signals.

- A **calibrated logistic regression** for interpretable win probabilities with honest OOF evaluation.
- A **defensively-adjusted line disparity metric** that controls for matchup confounding.
- A clear finding that **line balance does not strongly predict team success** ($r = -0.031$).

The model prioritizes **interpretability over marginal accuracy gains**: a domain expert can audit every step from raw data to final predictions. This transparency is the approach's greatest strength.

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